

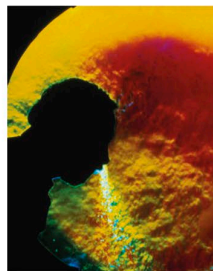
RESPIRATORY DISORDERS

MILLIONS OF MICROBES FLOAT IN THE AIR, AND EACH BREATH BRINGS THOSE PARTICLES INTO THE RESPIRATORY TRACT, HEIGHTENING THE RISK OF A RESPIRATORY INFECTION. OTHER TYPES OF RESPIRATORY DISORDER INCLUDE DAMAGE CAUSED BY ALLERGIES OR IRRITANTS, AND CANCERS.

COMMON COLD

The common cold is one of the most frequently experienced illnesses but also generally one of the less serious. At least 200 different and highly contagious types of virus can cause the problem. They spread in fluid that floats through air, in tiny droplets of mucus coughed or sneezed out by people with colds, and also in films of moisture transferred from person to person by close contact, such as shaking hands, or via shared objects, such as cups. Symptoms involve frequent sneezing, a runny nose, which at first runs with a clear, thin

discharge that may later become thicker and greenish-yellow, a headache, slightly raised temperature, and perhaps a sore throat, cough, and reddened eyes. Antibiotic drugs are ineffective as they do not work against viruses. Cold viruses change (mutate) so rapidly that even if antiviral drugs could be made to tackle existing strains, they would be ineffective against the new ones. Most cold remedies, such as decongestants and inhalants, treat the symptoms while the body's immune system attacks the invading microbes.

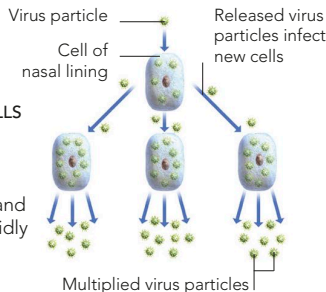


SPREADING INFECTION

Coughs and sneezes can rapidly spread common cold viruses by spraying them up to 3m (10 ft) in mucous droplets.

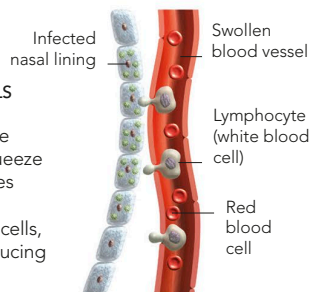
1 VIRUS INVADES CELLS

Virus particles in air land on and invade the cells lining the nose and throat. They rapidly replicate, killing their host cells.



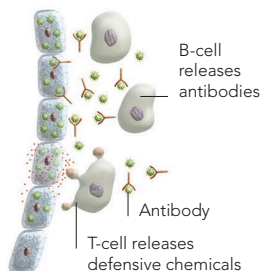
2 WHITE CELLS ARRIVE

Defensive white blood cells squeeze out of capillaries towards the infected lining cells, which are producing thin mucus.



3 ANTIBODY PRODUCTION

White blood cells known as B-cells produce antibodies, which immobilize the virus; other white blood cells destroy infected cells.



4 CLEARING UP

Other white blood cells called phagocytes engulf virus particles, damaged nasal lining cells, and other debris. The cold subsides.

